

Analysis of Modern Advances in Medical Segmentation for Heart Applications

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Abstract

Computer vision is a rapidly evolving field within artificial intelligence, driven by advancements and new implementations, specifically with deep-learning approaches. As the field continues to grow, new algorithms for Medical Segmentation are becoming more prevalent. This is useful for making accurate diagnoses, reliable identification of body parts and tissues, and providing medical professionals with more information to make treatment decisions. We will be completing a comprehensive study of four modern medical segmentation algorithms, explaining the novelty, loss functions, and overall architecture of each. Experimental results will be derived by looking at two differing heart medical segmentation datasets, which will be used for training and testing performance. Performance will be evaluated using standard segmentation metrics such as Dice Loss and Jaccard Index, testing binary segmentation and multi-class segmentation. Our findings provide insight into the strengths and current limitations of modern medical segmentation techniques. The code and datasets are available at <https://github.com/cbmusonda/ComputerVision.git>.

1. Introduction

Cardiovascular diseases account for nearly "18 million" deaths annually, making accurate cardiac image segmentation critical for diagnosis and treatment planning. How-

ever, developing robust deep learning segmentation models is hindered by the need for large volumes of expertly annotated medical images, which are time-consuming and expensive to obtain. Semi-supervised learning addresses this challenge by leveraging both limited labeled data and abundant unlabeled data, with recent methods employing techniques such as consistency regularization, pseudo-labeling, and contrastive learning. This paper presents a comprehensive comparative study of four state-of-the-art semi-supervised segmentation algorithms, Cross-View Mutual Learning (CML), Mutual Consistency Learning (MC-Net), Rectified Contrastive Pseudo Supervision (RCPS), and S4CVnet applied to cardiac imaging tasks. We evaluate these methods on the LA dataset for binary left atrium segmentation and the ACDC dataset for multi-class segmentation, examining performance under varying labeled data constraints (20% and 10%). Our findings, assessed using Dice Score, Jaccard Index, HD95, and ASD metrics, provide insights into the strengths and limitations of modern semi-supervised techniques for cardiac medical image segmentation.

2. Related Work

2.1. Semi-supervised Medical Segmentation

Applications of medical image segmentation extend beyond large organs such as the heart, and recent advances, specifically the RD-Net framework, have illustrated the development of semi-supervised algorithms that perform well with limited amounts of labeled images. The RD-Net framework

was introduced in a paper submission for CVRP 2026 titled “Boost the Inference with Co-training: A Depth-guided Mutual Learning Framework for Semi-supervised Medical Polyp Segmentation”, providing a state-of-the-art approach to semi-supervised learning using Mean Teacher[2]. Mean Teacher is a common technique used in semi-supervised learning where the unlabeled image data is larger than that of the labeled image data. There is a relationship between a Student and Teacher model in this technique, which allows for the modules to continue to learn based on image data. This learning is done via an input of both unlabeled and labeled image data into a Student Model, which has much more sporadic weight definitions after each iteration of training, and the Exponential Moving Average of the Student Model is sent to the Teacher Model for a more consistent way for the algorithm to learn. The Teacher Model creates soft targets that the Student Model can use to make their learning more accurate.

The RD-Net framework serves as a benchmark of semi-supervised learning algorithms by using existing methods with novel implementations to make the application easier to use with existing medical equipment and more accurate than previous models. There are two main implementations that RD-Net achieves, being that the depth image data used for training is only required for training and depth-guided patch augmentation. Both of these in action allow for the algorithm to work well with images that have ambiguous depths and difficult orientations, specifically for polyp detection applications. Being able to understand the success of this paper in the scope of semi-supervised medical segmentation has been critical in evaluating the algorithms covered in this paper from an informed lens.

2.2. Cross-Modal Mutual Learning

An important consideration in semi-supervised segmentation, particularly with Mean Teacher, is the mutual learning aspect, which is the exact interaction between the student and teacher models. Since many of the algorithms that will be analyzed later in the paper use similar techniques, it is important that we present a baseline understanding of what this looks like. Looking specifically at the implementation of mutual learning in the RD-Net framework, consistency learning, complementary knowledge learning, and auxiliary student network guidance were introduced. Consistency learning is a constraint placed upon unlabeled data, which helps the student networks used in the model have more predictable outputs. This is input in the algorithm as a new loss function based on the mean square error, added to the overall loss equation to be used for updating weights.

The engineers behind the algorithm needed a way to avoid premature convergence of the student model’s learning from the introduction of the previous constraint, which was combated by complementary knowledge learning. The

objective was to find a way to quantify the cross-modal relationship between the depth image student model and the RGB student model respectively. This is inherently an issue due to the depth and RGB images being different and having dissimilar features. The proposed solution was to ignore the similarities of the features and rather look at the output as a whole, which can then be separated into the binary segmentation classes of background and, in this case, polyp detected. The sample pairs of the output results in deeper learning, now being able to determine what is different and what is the same.

The final technique for improving learning involves the student model that trains on the depth images and the teacher model that outputs soft targets. The proposed solution involves the detected class from the output of the depth student model and teacher model, defining a threshold for background and polyp detection. If this set threshold distance was crossed, the teacher model output was switched out with the output of the student model. This protocol was executed every set number of pixels to allow for continuous learning.

Overall, many of these learning techniques are not unique to polyp segmentation, being discussed in the heart segmentation algorithms that this paper analyzes. The reasoning behind each method of improving learning is critical for understanding the architecture and novelties mentioned later, and can also serve as potential methods to improve existing algorithms.

3. Proposed Modern Architectures

3.1. CML Architecture

The first algorithm introduced in this paper is Cross-View Mutual Learning, which is a semi-supervised medical segmentation framework that introduces novel ways of learning across different subnets, similar to the implementation of RD-Net.

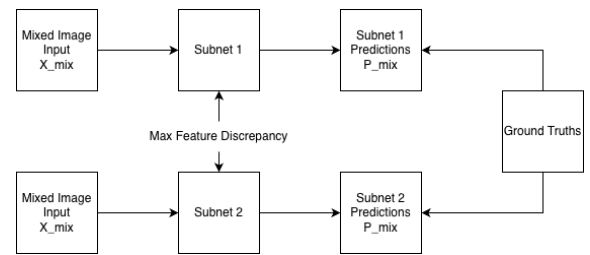


Figure 1. General Example Framework for Cross-View Mutual Learning

The framework comes from “Cross-View Mutual Learning for Semi-Supervised Medical Image Segmentation,” which was accepted to ACM Multimedia Conference in 2024[4]. The author intended to build off of the pre-existing

techniques, such as Mean Teacher, that use multiple subnets to produce consistent output information. There are two main novelties presented in this paper, with the first being conflict-based feature learning (CFL), which is a way that the algorithm can continue to learn without new image data by augmenting the image to have a different perspective or orientation, which then allows the network to learn distinct features. With the CFL established as a way to continue learning, the author’s next goal was to maximize the potential of this feature learning using cross-view mutual learning (CML), which simplifies the pseudo-label correction, only needing modifications to the inputs and supervisory signals.

3.2. CML Loss Function

Now that an overall understanding of the CML Architecture is established, we can show the equations that the authors used to develop their loss functions for gauging how well it performs and updating weight parameters to improve the results. The first loss equation is the feature-level discrepancy loss below:

$$\mathcal{L}_{dis,i}^{\alpha} = 1 + \frac{f_i^{\alpha} \cdot \bar{f}_{(1-i)}^{\alpha}}{\|f_i^{\alpha}\| \times \|\bar{f}_{(1-i)}^{\alpha}\|} \quad (1)$$

Where f_i^{α} is the output from the decoder and $\bar{f}_{(1-i)}^{\alpha}$ is the copy of a feature output from the decoder without gradients. Since this framework has two different subnets, the discrepancy loss is to be written as the sum of the feature-level losses with notation of $\mathcal{L}_{dis}^{\alpha}$.

Now that a way for the features learned from the algorithm has been quantified in a way to improve learning through the discrepancy loss, the authors begin deriving their loss equation for the supervised learning done with the labeled image data.

$$\mathcal{L}_{sup,i}^l(P_i^l, Y^l) = \frac{1}{2} (\mathcal{L}_{ce}(P_i^l, Y^l) + \mathcal{L}_{dice}(P_i^l, Y^l)) \quad (2)$$

Above is the definition of the supervised loss with a notation of $\mathcal{L}_{sup,i}^l(P_i^l, Y^l)$, where it is a function of the probability output from the respective subnet, and the ground truth (labeled data). The important takeaway from the supervised loss is that the authors chose to derive this as the summation of the cross-entropy and dice losses. Cross-entropy loss describes the difference between a given probability distribution and the expected distribution, while dice loss measures the overlap between the ground truth and the output. $\mathcal{L}_{sup}^l$ is the summation of both supervised loss equations from each subnet.

The final component for the CFL total loss is the unlabeled image loss, which is derived using the supervised loss equation mentioned above with differing inputs based on the probability outputs of unlabeled data and a pseudo-label generated from taking the argmax of this said probability.

This results in the following equation for unlabeled loss:

$$\mathcal{L}_{sup}^u = \mathcal{L}_{sup,0}^u(P_0^u, \tilde{Y}_1^u) + \mathcal{L}_{sup,1}^u(P_1^u, \tilde{Y}_0^u) \quad (3)$$

A significant distinction of the unlabeled supervised loss is that the inputs of probability and pseudo-labels will always be from different subnets. This means that constructing the loss for subnet 0 takes the pseudo-labels generated from subnet 1 and its own probability output, and vice versa. The total loss for the proposed CFL method is:

$$\mathcal{L} = \mathcal{L}_{sup}^l + \lambda_1 \mathcal{L}_{sup}^u + \lambda_2 \mathcal{L}_{dis} \quad (4)$$

Note: λ_1 and λ_2 are weights for the corresponding loss they are attached to and $\mathcal{L}_{dis}$ is the labeled and unlabeled feature discrepancy loss.

With all of the contributing factors towards a loss function for CFL established, the authors then create a way to establish this via cross-view mutual learning, which involves altering input image data for continued learning. The input and label data for training on labeled images is altered using CutMix which augments the image. From there two labeled inputs are taken to derive the new mixed input below, with their ground truths used for the mixed output equation.

$$X_{mix}^l = X_u^l \odot \mathcal{M} + X_b^l \odot (1 - \mathcal{M}) \quad (5)$$

$$Y_{mix}^l = Y_u^l \odot \mathcal{M} + Y_b^l \odot (1 - \mathcal{M}) \quad (6)$$

The new mixed input and mask values allow for a new loss specific to CML to form using this new data, which is a summation of label supervised loss for both subnets, with inputs of mixed pseudo-labels and probability outputs.

$$\mathcal{L}_{sup}^l = \mathcal{L}_{sup,0}^l(P_{mix,0}^l, Y_{mix}^l) + \mathcal{L}_{sup,1}^l(P_{mix,1}^l, Y_{mix}^l) \quad (7)$$

Similar principles are applied to the unlabeled loss with the following mixed input and pseudo-label equations.

$$X_{mix}^u = X_p^u \odot \mathcal{M} + X_q^u \odot (1 - \mathcal{M}), \quad (8)$$

$$Y_{mix,i}^u = \hat{Y}_{p,(1-i)}^u \odot \mathcal{M} + \hat{Y}_{q,i}^u \odot (1 - \mathcal{M}), \quad (9)$$

From this the unsupervised loss specific to CML is defined below:

$$\mathcal{L}_{sup}^u = \mathcal{L}_{sup,0}^u(P_{mix,0}^u, Y_{mix,0}^u) + \mathcal{L}_{sup,1}^u(P_{mix,1}^u, Y_{mix,1}^u) \quad (10)$$

With both the labeled and unlabeled losses quantifiable, the summation of the new labeled loss with a λ_2 weight for discrepancy loss can define the pre-training loss function.

3.3. MC-Net Architecture

The next paper we chose to evaluate for its novel implementation of mutual consistency was "Mutual Consistency Learning for Semi-supervised Medical Image Segmentation", which was submitted to Medical Image Analysis in

2022[5]. Similar to all of the papers that will be covered in this analysis, the authors wanted to address the known issue of unpredictable and sporadic classification in semi-supervised applications that use large amounts of unlabeled data. This architecture introduces techniques based

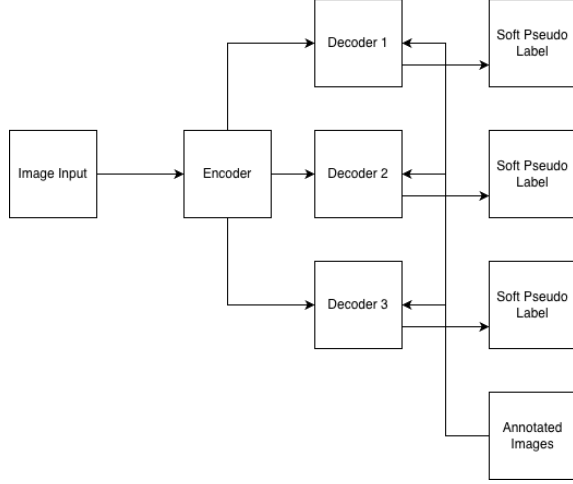


Figure 2. General Example Framework for Mutual Consistency Learning

on probabilities of multiple decoder networks (V-net or U-net) to achieve more accurate results around spatially ambiguous areas of a given image, which is a critical task in medical segmentation. The main premise of the framework is to take the probabilistic outcomes of a given decoder and apply mutual consistency with pseudo-label information from different decoders in the architecture. There are two main contributions the authors note, being the mutual consistency previously discussed, and a way to force low-entropy predictions for the unlabeled data to get an overall better performance compared to other modern models. The model takes inspiration from multi-tasking learning and uncertainty estimation techniques to develop their model. Multi-task learning is a general way to learn cross-task feature representation, which can typically be done by implementing some form of pre-training (which the CML does). For MC-Net, there was no need to complete any auxiliary tasks other than the actual segmentation model itself to make their model work well. Uncertainty estimation was used by the authors to pre-define their sub-models, which allows for an efficient model uncertainty calculation, which other existing models complete with larger computational costs.

3.4. MC-Net Loss Function

Now that the general functions that MC-Net strives to improve upon we can begin to look at the loss functions and accompanying equations provided by the authors. The first equations that must be defined are the $f_{\theta_{sub}}$ and μ_x , where

the first is the a subnet generated from sampled of the full network and the second is the pixel uncertainty equation that will be used for making the unlabeled data useful later.

$$f_{\theta_{sub}} = Sampling(f_{\theta})$$

$$\mu_x = D[p(y_{pred}|x; \theta_{sub}^1), \dots, p(y_{pred}|x; \theta_{sub}^n)] \quad (1)$$

Speaking more on the pixel uncertainty equation, the way this functions is by multiplying the discrepancy between a set amount of outputs D , with the sum of probabilities that a label predictions will align with a given input from a set subnet. The subnet equation can be broken down into a more meaning $f_{\theta_{sub}^i}$ which will put the subnet in terms of a single encoder for the model and one of n decoders, since this is how the author intends to define their subnets, which can make finding the uncertainty based on probabilities of a given subnet more accurate.

$$f_{\theta_{sub}^i} = f_{\theta_e} \boxplus f_{\theta_d^i}, \quad i \in \{1, \dots, n\} \quad (11)$$

It should be noted that the authors used a value of $n = 3$ to implement their framework, which was a choice driven by avoiding complexity while still having enough subnets (need one decoder for each) to get effective results.

Going into the training of this framework and the loss functions that allow for the code to evaluate its own performance, the main novel loss introduced is L_{mc} which is the contribution from the mutual consistency, which is used for unlabeled and labeled data. The equation is as follows:

$$L_{mc} = \sum_{i,j=1 \& i \neq j}^n D[p^*(y_{pred}^*|x; \theta_{sub}^i), p(y_{pred}|x; \theta_{sub}^j)] \quad (12)$$

This equation is fairly similar to what was previously discussed, where the prediction from a given subnet is compared to this new prediction, which is a pseudo-label soft target generated through the following:

$$p^*(y_{pred}^*|x; \theta) = \frac{p(y_{pred}|x; \theta)^{1/T}}{p(y_{pred}|x; \theta)^{1/T} + (1 - p(y_{pred}|x; \theta))^{1/T}} \quad (13)$$

To break this equation down, the $p(y_{pred}|x; \theta)$ is an probability map, where when put in the configuration above, with an accompanying $1/T$, the end results is a soft target. The T corresponds to a temperature variable, where temperature is typically a value used to control the accuracy of the model.

The final loss can be expressed in a similar fashion to CML where hyper parameters λ and β are included with corresponding L_{seg} and L_{mc} .

$$Loss = \lambda \times \sum_{i=1}^n L_{seg}(p(y_{pred}|x_i; \theta_{sub}^i), y_i) + \beta \times L_{mc} \quad (14)$$

3.5. RCPS Architecture

The third algorithm introduced in this paper is Rectified Contrastive Pseudo Supervision (RCPS), which is a semi-supervised medical image segmentation framework designed to address two key limitations of pseudo-label based learning: the noisy nature of pseudo labels and the lack of explicit supervision in the feature space. The method, presented in the paper "RCPS: Rectified Contrastive Pseudo Supervision for Semi-Supervised Medical Image Segmentation" published in Medical Image Analysis in 2022[7], builds upon a U-Net backbone and introduces additional mechanisms that operate both in the label space and in the feature space.

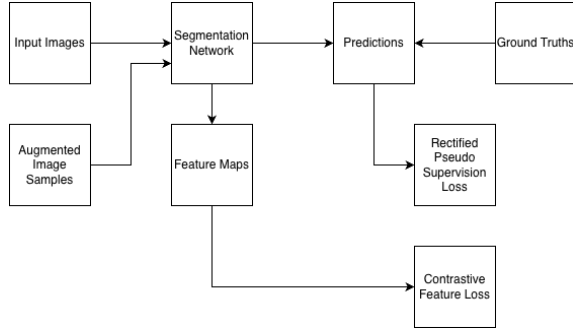


Figure 3. General Example Framework for Rectified Contrastive Pseudo Supervision

Given an input image x , RCPS generates two augmented views x_{ϕ_1} and x_{ϕ_2} using intensity-based transformations such as brightness and contrast variation, as well as random Gaussian noise. These augmentations simulate appearance variability commonly encountered in medical imaging due to differences in acquisition protocols and equipment. The segmentation network produces predictions for the original input and both augmented views, denoted as $\hat{y}$, $\hat{y}_{\phi_1}$, and $\hat{y}_{\phi_2}$, respectively.

To enable feature-level supervision, RCPS extracts intermediate decoder features and maps them into a lower-dimensional embedding space using a projection head. These projected features are used to perform voxel-level contrastive learning, where features corresponding to the same semantic class are encouraged to be closer, while features from different classes are pushed apart. In addition, an auxiliary augmented input x^- is used to supply negative samples for contrastive learning.

The architectural novelty of RCPS lies in its dual strategy: rectified pseudo supervision improves the reliability of pseudo labels through uncertainty-aware weighting and consistency regularization, while bidirectional voxel-level contrastive learning enforces class separability directly in the feature space. This combination enables RCPS to outperform methods that rely solely on label-space supervi-

sion.

3.6. RCPS Loss Function

For labeled data, RCPS employs a standard supervised segmentation loss composed of cross-entropy loss and Dice loss:

$$\mathcal{L}_{seg}(\hat{y}, y) = \mathcal{L}_{ce}(\hat{y}, y) + \mathcal{L}_{Dice}(\hat{y}, y) \quad (15)$$

where y denotes the ground truth segmentation mask.

For unlabeled data, pseudo supervision is applied by encouraging consistency between predictions from the original image and augmented views. To sharpen pseudo labels and enforce the low-density separation assumption, logits are scaled using a temperature parameter T :

$$\mathcal{L}_p(\hat{y}_{\phi_i}, \hat{y}) = \mathcal{L}_{ce}(\hat{y}_{\phi_i}, \sigma(z/T)) \quad (16)$$

where z represents the logits of the original prediction and $\sigma(\cdot)$ denotes the softmax function.

To mitigate the influence of noisy pseudo labels, RCPS introduces uncertainty rectification based on KL-divergence between predictions:

$$D_{kl}(\hat{y}_{\phi_i}, \hat{y}) = \hat{y} \log \left(\frac{\hat{y}}{\hat{y}_{\phi_i}} \right) \quad (17)$$

The rectified pseudo supervision loss is then defined as:

$$\mathcal{L}_{urp} = e^{-D_{kl}} \mathcal{L}_p + D_{kl} \quad (18)$$

Consistency regularization between the two augmented views is enforced using cosine similarity:

$$\mathcal{L}_{cr} = 1 - \frac{\hat{y}_{\phi_1} \cdot \hat{y}_{\phi_2}}{\|\hat{y}_{\phi_1}\|_2 \|\hat{y}_{\phi_2}\|_2} \quad (19)$$

Combining these components yields the rectified pseudo supervision loss:

$$\mathcal{L}_{rp} = \mathcal{L}_{urp}(\phi_1) + \mathcal{L}_{urp}(\phi_2) + \mathcal{L}_{cr} \quad (20)$$

To enforce feature-space consistency, RCPS introduces bidirectional voxel contrastive learning based on the InfoNCE loss. Given a pair of corresponding voxel embeddings ψ_1 and ψ_2 , the bidirectional contrastive loss is defined as:

$$\mathcal{L}_{bc} = \mathcal{L}_c(\psi_1, \psi_2) + \mathcal{L}_c(\psi_2, \psi_1) \quad (21)$$

where confident negative sampling is employed to reduce false-negative pairs.

The final objectives are defined as:

Unlabeled data:

$$\ell_{unsup} = \alpha \mathcal{L}_{rp} + \beta \mathcal{L}_{bc} \quad (22)$$

Labeled data:

$$\ell_{sup} = \mathcal{L}_{seg} + \alpha \mathcal{L}_{rp} + \beta \mathcal{L}_{bc} \quad (23)$$

where α and β control the contribution of pseudo supervision and contrastive learning.

3.7. S4CVnet Architecture

The fourth algorithm covered in this analysis is S4CVnet (Semi-Supervised Semantic Segmentation with CNN and ViT), which is a semi-supervised medical image segmentation framework that explicitly leverages the complementary strengths of Convolutional Neural Networks (CNNs) and Vision Transformers (ViTs). The method, presented in the paper "When CNN Meet with ViT: Towards Semi-Supervised Learning for Multi-Class Medical Image Semantic Segmentation" published in arXiv in 2024[3], is motivated by the observation that CNNs excel at capturing local spatial details, while ViTs are more effective at modeling long-range global dependencies.

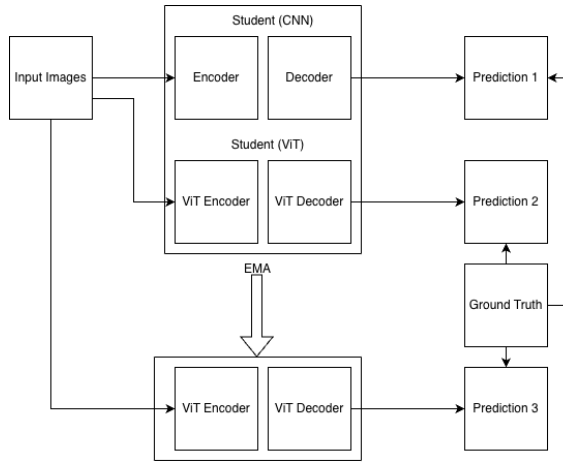


Figure 4. General Example Framework for S4CVnet with CNN and ViT

The framework consists of two main components: a feature-learning module and a guidance module. The feature-learning module contains two heterogeneous segmentation networks: a CNN-based U-Net and a ViT-based U-Net. These two networks are trained jointly using a dual-view co-training strategy, where each network provides pseudo labels to supervise the other. Because the two architectures extract fundamentally different representations, their predictions are naturally diverse, reducing the risk of confirmation bias commonly seen in homogeneous co-training frameworks.

The guidance module introduces an additional ViT-based network whose parameters are updated using an exponential moving average (EMA) of the feature-learning ViT parameters. This EMA model acts as a stable teacher that generates higher-quality pseudo labels for consistency supervision. During training, perturbations are applied to the feature-learning networks, while the guidance network remains stable due to EMA updates, providing robust supervisory signals under noisy conditions.

3.8. S4CVnet Loss Function

For labeled data, each feature-learning network is optimized using a supervised segmentation loss composed of cross-entropy and Dice loss:

$$\mathcal{L}_{sup} = \frac{1}{2} (\mathcal{L}_{ce}(Y_{gt}, f(X)) + \mathcal{L}_{Dice}(Y_{gt}, f(X))) \quad (24)$$

where $f(X)$ represents either the CNN or ViT segmentation network.

To exploit unlabeled data, S4CVnet introduces cross pseudo supervision between the CNN and ViT networks. The CNN is supervised using pseudo labels generated by the ViT, and vice versa:

$$\mathcal{L}_{semi1} = \mathcal{L}_{ce}(\arg\max(f_{CNN}(X)), f_{ViT}(X)) \quad (25)$$

$$\mathcal{L}_{semi2} = \mathcal{L}_{ce}(\arg\max(f_{ViT}(X)), f_{CNN}(X)) \quad (26)$$

In addition, consistency-aware supervision is applied using pseudo labels from the EMA-based guidance network:

$$\mathcal{L}_{semi3} = \mathcal{L}_{ce}(\arg\max(f_G(X)), f_{ViT}(X)) \quad (27)$$

$$\mathcal{L}_{semi4} = \mathcal{L}_{ce}(\arg\max(f_G(X)), f_{CNN}(X)) \quad (28)$$

The overall training objective combines supervised and semi-supervised losses:

$$\mathcal{L} = \mathcal{L}_{sup1} + \mathcal{L}_{sup2} + \lambda_1(\mathcal{L}_{semi1} + \mathcal{L}_{semi2}) + \lambda_2(\mathcal{L}_{semi3} + \mathcal{L}_{semi4}) \quad (29)$$

where λ_1 and λ_2 are time-dependent weighting coefficients that gradually increase during training to reduce the impact of unreliable pseudo labels in early iterations.

4. Experiments

4.1. Dataset and Evaluation Metrics

The four state-of-the-art algorithms analyzed in this paper were evaluated by us using two datasets for medical segmentation, specifically for the heart. The first being the LA [6] dataset which is comprised of 100 3D MRI scans with corresponding ground truths. This dataset is used for evaluating how well each model performs binary segmentation tasks in which 0 corresponds to background and 1 corresponds to Left Atrium. The isotropic resolution of these scans is $0.625 \times 0.625 \times 0.625 \text{ mm}^3$. The second dataset is ACDC [1] which is comprised of 100 2D MRI images, with three separate classes annotated being Left Ventricle, Right Ventricle, and Myocardium. This dataset was used for evaluating how well each model can perform for multi-class heart segmentation, whether metrics improve or decline with this new dilemma.

There were four main metrics used for evaluating the performance of each algorithm, with the first being Dice

Score. The Dice Score is a way to measure the overlap between a predicted mask P and the corresponding ground truth G , calculated by the ratio of correct pixels classified divided by the total area as follows:

$$\text{Dice} = \frac{2|P \cap G|}{|P| + |G|} \quad (30)$$

The Jaccard Index also referred to as IoU is quantifying the same thing that the Dice Score does but from a different mathematical approach:

$$\text{IoU} = \frac{|P \cap G|}{|P \cup G|} \quad (31)$$

As the name suggests, the intersection of the predicted mask and the ground truth is divided by the union, which is all correctly predicted pixels over the total pixels in the ground truth.

The Hausdorff Distance 95th Percentile (HD95) is a metric that measures a distance between predicted and true boundaries, avoiding absolute max distances, only looking up to the 95th percentile of distances. This works by generating a list of all distances and ignoring the extreme 5 percent outliers, taking the worst case scenario values from this list in both directions:

$$d_{95}(A, B) = X_{95} \left\{ \min_{a \in A} d(a, b) \right\}_{b \in B} \quad (32)$$

$$\text{HD95}(A, B) = \max \{d_{95}(A, B), d_{95}(B, A)\} \quad (33)$$

The final metric utilized for this study is the Average Surface Density (ASD), which is a way to quantify the mismatch between a prediction mask and the ground truth by calculating the mean surface distance from positions on one surface to that of another.

$$\text{ASD}(A, B) = \frac{1}{|A|} \sum_{a \in A} \min_{b \in B} d(a, b) \quad (34)$$

The equation above describes the nature of this metric, being that for every point of a predicted boundary the closest distance to the ground truth will be calculated.

4.2. Implementation Details

All of the experiments from this study were completed on Google Colab using a NVIDIA A100 Tensor Core GPU available to us through a student account. For the experiments with the LA and ACDC datasets we wanted to observe the performance each model had, with the metrics mentioned above as benchmarks for how well each one did. We also wanted to introduce the % of labeled data used for training as variable to see what differences, if any, arose when switching from 20% to 10% labeled data. For the LA dataset, all of the models completed 15,000 iterations for

training, where 112 x 112 x 80 patch sizes were randomly extracted from the dataset for use. For the ACDC dataset, all of the models completed 30,000 iterations for training, where 256 x 256 patches were randomly extracted. All of the models had a learning rate of 0.01.

4.3. Results

After successfully running training and testing for the four different algorithms evaluated in this paper, we were able to put the results in tabular form evident in Tables 1 and 2, corresponding to LA and ACDC, respectively.

Table 1. Results from LA Dataset Experiment for 10% and 20% Labeled Images

LA	Dice (%)	Jaccard (%)	95HD (px)	ASD (px)
CML	90.08	82.04	5.91	1.77
MC-Net	88.96	80.27	8.75	1.66
RCPS	65.93	51.88	31.39	9.56
S4CVnet	49.99	37.61	24.66	7.54
CML	90.83	83.26	5.91	1.66
MC-Net	90.67	83.04	6.15	1.79
RCPS	79.09	67.73	12.83	3.71
S4CVnet	55.29	42.13	23.4	7.41

Table 2. Results from ACDC Dataset Experiment for 10% and 20% Labeled Images

ACDC	Dice (%)	Jaccard (%)	95HD (px)	ASD (px)
CML	88.89	80.66	2.09	0.61
MC-Net	86.19	76.75	10.94	2.74
RCPS	62.8	80.56	92.37	39.86
S4CVnet	76.93	66.77	10.07	2.61
CML	89.22	81.21	3.41	0.86
MC-Net	87.77	79.25	5.54	1.64
RCPS	40.86	50.51	97.5	44.44
S4CVnet	87.32	78.67	7.69	2.22

The best results were put in bold, and the first half of each table is 10% labeled data, while the bottom is for the 20% results. Overall, we noticed that the CML algorithm was consistently getting the highest performance with the Dice values being in the 90s and high 89.9s, which is impressive. There were some values that seemed too far off from what was to be conventionally expected from them, whether that be very low dice and Jaccard or very high 95HD and ASD. We noticed that it was interesting to look at what areas the other three algorithms performed well in, as the 2nd best performance was not usually the same from metric to metric. What was also interesting to observe was

the large jump in performance with the introduction of additional labeled information.

5. Conclusion

In this work, we completed a comprehensive evaluation of four different state-of-the-art medical segmentation algorithms. We established the existing background of this specific application of computer vision and what our motivation for this paper was. With this set, we found papers that aligned with heart medical segmentation and analyzed each framework and its accompanying loss functions. The two main experiments set out were to investigate the performance of multiple existing architectures when trained with differing amounts of labeled data, along with seeing how performance was affected when multiple classes are introduced. Our experiments showed that CML had the best overall performance compared to the other frameworks, with testing and training done via the ACDC and LA datasets. This work served as a checkpoint for what existing methods were out there for medical segmentation, explaining each architecture and seeing what works the best as of now. As time goes on there will be new papers with innovate solutions that will improve performance. For future work to be done, it would be more meaningful to include more models and a wider variety of datasets to get a better understanding of the success engineers have had for semi-supervised techniques.

Andrew and Changwe split up the training and testing of the different architectures (2 each). Andrew worked on the MC-Net and CML training and testing, while Changwe worked on the RCPS and S4CVnet training and testing. All three members worked on the report, Changwe working on the Introduction and Experiments section. Manav focused on writing out the technical details of RCPS and S4CVnet. Andrew focused on the MC-Net and CML technical explanation, along with the abstract, related work and the experiments. All three members worked on the presentation slides.

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